SmartTAR - STAR v1.0

Detailed summary: what it is, what it can do, and what it is for

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In short: SmartTAR - STAR is a Windows PowerShell GUI archiver that uses the system tar.exe / bsdtar engine, while adding its own transparent STAR archive format. It combines an outer TAR container, an internal manifest, block-based storage, SHA-256 verification, and smart compression planning.

1. What SmartTAR - STAR is

SmartTAR - STAR is a desktop archiving tool for Windows, written in PowerShell and equipped with a graphical user interface. It is not only a simple wrapper around tar.exe. The program creates its own .star archive container, where the archived data is stored as a set of internal blocks and described by a manifest containing metadata.

From a practical point of view, it is designed for users who want a simple GUI but also appreciate a technically readable, verifiable, and partially recoverable archive structure. The foundation remains the TAR technology available directly in Windows, while SmartTAR adds planning, checksums, reports, and a safer working workflow.

2. What it is for

The program is intended for archiving files and folders on Windows. It is especially useful where simply packing data is not enough, and where archive structure, verification, and partial recovery matter.

- Creating a single .star archive from files or folders.
- Building verifiable archives with SHA-256 checksums for individual blocks.
- Archiving mixed data where different file types may benefit from different compression behavior.
- Using a block-based structure instead of one monolithic compressed stream.
- Verifying already created archives through a dedicated VERIFY action.
- Attempting to recover usable data from partially damaged archives using Salvage mode.

3. What the program can do

SmartTAR works in three main modes: compression, extraction, and verification. In addition, it provides an optional salvage mode for extraction from problematic archives.

Area	Description
Compression	Selects an input file or folder, prepares a working structure, divides content into blocks according to the selected mode, and creates the resulting .star archive.
Extraction	Unpacks the outer STAR container, reads the manifest, and extracts the internal blocks into the target location.
Verification	Checks archive blocks, their readability, and their SHA-256 checksums. The result is written to a report.
Salvage mode	When an archive is damaged, the program attempts to extract usable blocks and skip blocks that fail.
Worker mode	Long operations run in a hidden worker process, keeping the GUI more responsive and transferring status through temporary files.

4. How the STAR format works

The resulting .star file is an outer TAR container. Inside it are metadata and compression blocks. This means the archive is not just an opaque binary file without context; it has a readable internal logic.

- manifest.json contains information about the archive, compression mode, source name, blocks, and checksums.
- The blocks/ folder contains individual blocks, such as uncompressed TAR blocks or compressed tar.xz, tar.gz, tar.bz2, or tar.zst blocks depending on system capabilities.
- Each block has its own metadata and SHA-256 hash, allowing the archive to be verified block by block.
- The block structure increases the chance of partial recovery if the entire archive is not damaged.

5. Compression modes

Mode	Recommended use
Hybrid	Default general-purpose option. The program groups files by detected data type and uses suitable compression methods for the groups.
Smart	Creates blocks based on detected data types. Useful when you want a clearer internal archive structure.
Solid	Creates one larger compressed block using an automatically selected method. Simple and often efficient for compression.
Smart XZ	Uses grouped XZ blocks. Useful when strong compression and stable XZ staging behavior are preferred.
Store	Stores data without compression in TAR blocks. Useful for speed, testing, or already compressed data.

6. Key strengths

- More transparent archive model: the archive contains a manifest and blocks, not just an anonymous data stream.
- Verifiability: SHA-256 block checks help detect corruption.
- Practical GUI: users do not need to type tar commands manually.
- Responsiveness: long-running operations use a worker mechanism, allowing the interface to display ongoing status.
- Flexibility: multiple compression modes allow the user to balance speed, archive size, and internal structure.
- Recovery potential: salvage mode is useful when an archive is partially damaged but some blocks remain usable.

7. Typical workflow

During compression, the user selects a file or folder, chooses the target archive and compression mode. SmartTAR then prepares a safe temporary workspace, detects the capabilities of the system tar.exe, plans groups, creates blocks, builds the manifest, and finally produces the STAR archive.

During extraction, the program first unpacks the outer container into a temporary workspace, reads the manifest, checks path safety, and extracts blocks into the target location. During verification, instead of extracting data, it checks block readability and hashes, then creates a report.

8. Who it is suitable for

- Windows users who want a simple archiver without installing a complex toolchain.
- Technical users who appreciate manifests, block structure, and verification.
- Archiving projects, folders, work data, and mixed content.
- Scenarios where integrity checks and the ability to recover at least part of the data are important.

9. Limitations and practical notes

- SmartTAR relies on the system tar.exe / bsdtar available in Windows, so specific compression support depends on the system.
- The .star format is TAR-based, but its internal logic is specific to SmartTAR.
- For full verification, salvage extraction, and correct manifest handling, SmartTAR should be used.
- For very long paths or unusual characters, a test archive and verification run are recommended.

10. One-sentence summary

SmartTAR - STAR v1.0 is a practical Windows GUI archiver that turns the standard tar.exe engine into a smarter tool with its own .star container, block-based structure, SHA-256 verification, compression modes, and partial data recovery support.

Document prepared for the SmartTAR - STAR v1.0 release.